Project Proposal

Road Rescue : On road assistance companion

Course Name: Junior Design

Course Initial: CSE299

Section: 09

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Table Contents:

Road Rescue : On road assistance companion

Abstract:

Road Rescue is a mobile application designed to serve as the ultimate travel advisor for users. It offers on-road assistance, real-time updates, and essential travel features to ensure a safe and convenient journey.

Problem:

Road conditions can be unpredictable, and vehicle breakdowns or emergencies can occur at any time. In such circumstances, finding timely assistance and reliable information can be challenging. There is a need for a comprehensive and user-friendly app that can provide immediate on-road assistance, connect users with necessary services, and offer useful travel-related features.

Introduction:

Road Rescue is a mobile application designed to serve as a trustworthy on-road assistance companion for the users. The app is developed with the primary objective to provide prompt and reliable support during vehicle breakdowns and emergencies. Moreover, Road Rescue goes beyond conventional assistance, offering a range of features to enhance users with roadside aid, facilitate informed decision-making, and promote smooth travel experiences.

Features:

1. Basic car repair tutorial step. (After accessing this feature, user can have the basic idea regarding car repair)
2. Chat bot. (This feature will guide the user if any assistance is needed.)
3. Traffic and weather updates. (Through this feature, user can have the current traffic and weather updates)
4. GPS to track current location. (For emergency cases, user can share current location to his/her emergency contacts)
5. SOS signal to pre-selected contacts. (Can activate an SOS signal that instantly notifies pre-selected contacts, seeking immediate help.)
6. Emergency request to nearby workshop (Allows drivers to send an emergency request to nearby workshops, connecting them with skilled mechanics for rapid assistance.)
7. Find a hotel. (If user needs any accommodation, they can find it through this features)
8. Hospital Recommendation. (According to user budget and location, they can have recommended doctor, test, admission and ambulance list)
9. Carpooling support. (Allows users to find and connect with other travellers heading in the same direction, promoting eco-friendly travel and cost-sharing.)
10. Nearby Gas Station. (User can have the list of nearby gas stations based on their current location)

Target Audience and Market:

When a system is under pre-development, understanding the market and identifying the target audience is of utmost importance. Our system offers a range of services that are not limited to any specific group of people. Any person can access our system to avail its services. Since there is no gender-based restrictions, we can assume that a substantial number of users can access our system.

Here's a breakdown of the diverse audience that our app aims to satisfy:

- Individuals who seek reliable on-road assistance and information.
- Travellers who wish to stay informed about traffic and weather conditions.
- Those who prioritize safety and require prompt assistance during emergency situations.
- Tourists and travellers looking for convenient hotel and hospital locators during their journeys.

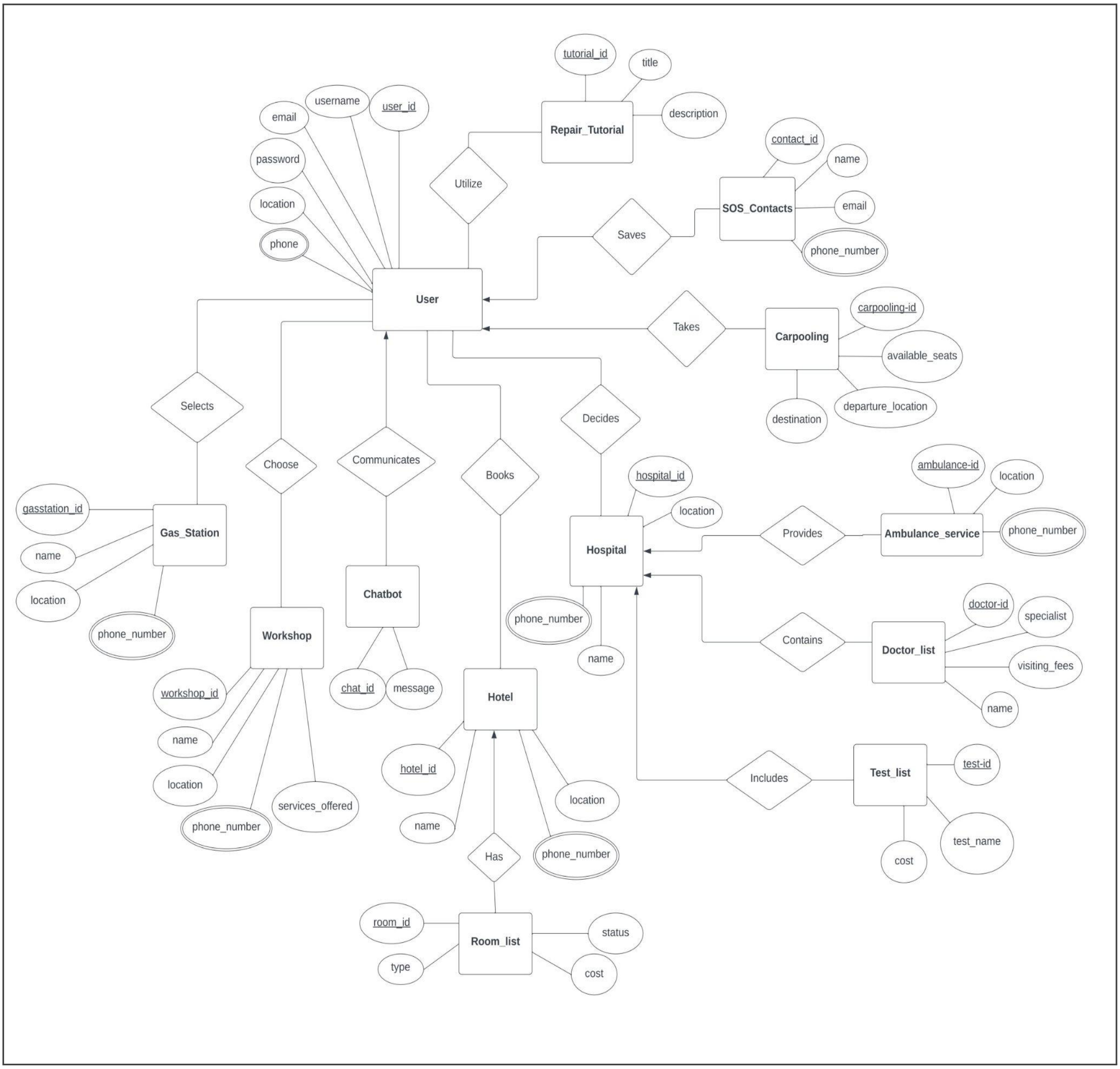
Technology:

Platform: Android (Mobile)

Database: XAMPP.

Language/Framework: Java (Spring)

ER Diagram:



Link :

https://lucid.app/lucidchart/1877ebfd-dc8a-4faa-be46-74cbec79d7eb/edit?viewport_loc=-146%2C225%2C2484%2C1050%2C0_0&invitationId=inv_42e4411a-37d3-4db9-86c1-526c0976f128

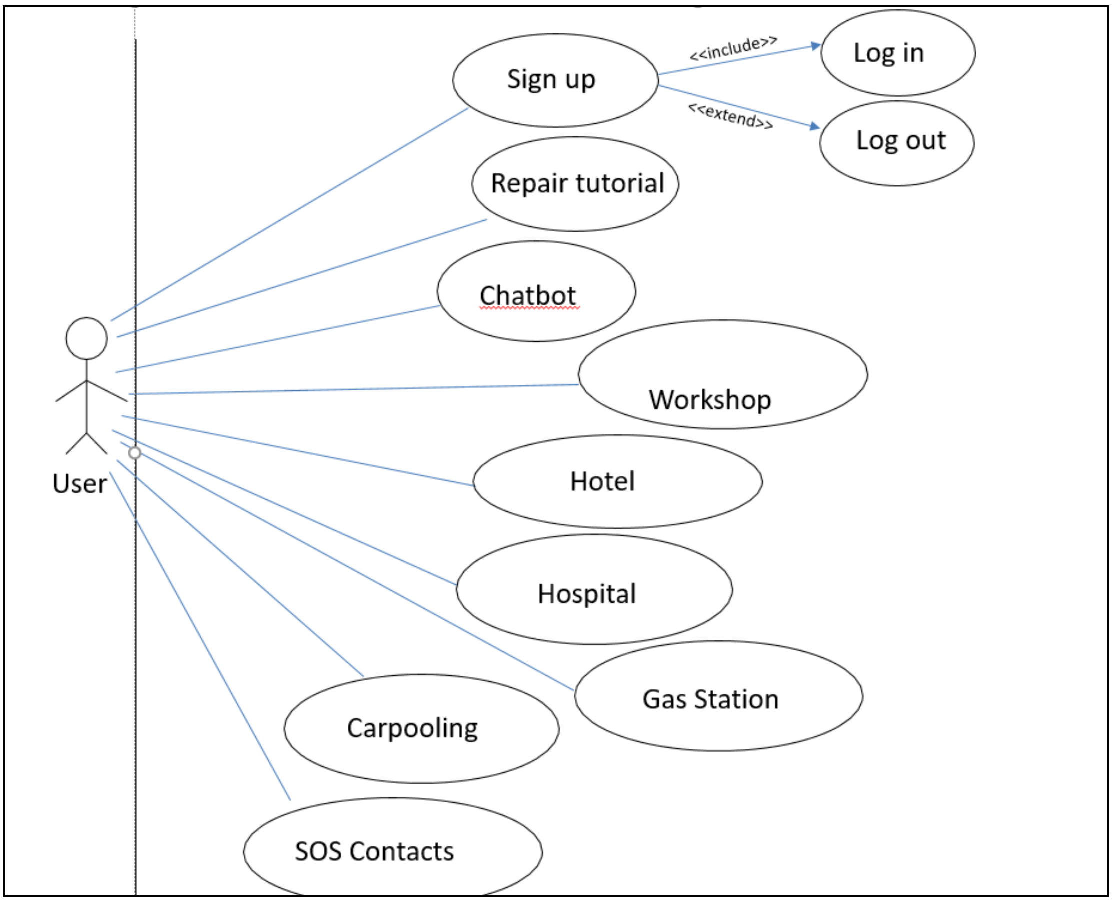
Relation schema

1. User (user_id, username, email, phone, password, location)
2. Repair_Tutorial (tutorial_id, title, description)
3. Utilize (user_id, tutorial_id)
4. Chatbot (chat_id, user_id, message)
5. Workshop (workshop_id, name, location, phone_number, services_offered)
6. Choose (workshop_id, user_id)
7. Hotel (hotel_id, name, location, phone_number)
8. Room_list (room_id, hotel_id, type, status, cost)
9. Books (user_id, hotel_id)
10. Hospital (hospital_id, name, location, phone_number)
11. Ambulance_service (ambulance_id, hospital_id, location, phone_number)
12. Test_list (test_id, hospital_id, test_name, cost)
13. Doctor_list (doctor_id, hospital_id, name, specialist, visiting_fees)
14. Decides (user_id, hospital_id)
15. Gas_Station (gasstation_id, name, location, phone_number)
16. Selects (gasstation_id, user_id)
17. SOS_contacts (contact_id, user_id, name, phone_number, email)
18. Carpooling (carpooling_id, user-id, departure_location, destination, available_seats)

Note:

Foreign Key

Use Case Diagram:



Timeline Chart:

PROJECT NAME	PROJECT DURATION	PROJECT START DATE	PROJECT END DATE
Road Rescue : On road assistance companion	13 weeks	July 24,2023	October 30,2023

Task ID	Task Description/ Person	Duration (Days)	Start Date	End Date	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11	WEEK 12	WEEK 13	WEEK 13
1	Project Idea	7	Jul 24, 2023	Jul 31, 2023														
2	Project Proposal	7	Jul 31, 2023	Aug 07, 2023														
3	User signup, login, repair UI	7	Aug 07, 2023	Aug 14, 2023														
4	Home page UI, Chatbot	7	Aug 14, 2023	Aug 21, 2023														
5	Hotel, Nearby workshop	14	Aug 21, 2023	Sep 04, 2023														
6	SOS signal, Nearby gas station	7	Sep 04, 2023	Sep 11, 2023														
7	Hospital recommendation	21	Sep 11, 2023	Oct 02, 2023														
8	Weather & Traffic updates, GPS	14	Oct 02, 2023	Oct 16, 2023														
9	Carpooling	7	Oct 16, 2023	Oct 23, 2023														
10	testing, Q/A	7	Oct 23, 2023	Oct 30, 2023														

Cost Analysis:

Constructive Cost Model:

Project Type : Organic

Coefficient$_{<\text{Effort Factor}>}$: 2.4 [P=1.05; T=0.38]

SLOC : 7500 lines (Source line of code)

Person Months, PM : $2.4 * (7500/1000)^{1.05} = 19.91$

Dev, Time, DM : $2.50 * 19.91^{0.38} = 7.79$ = 8 months = 1408 working hour
(22 days per month and 8 hour per day)

Required people, ST: PM/DM = 3 people

BUDGETING:

Developer Salary in 8 months

(Per Developer Salary Per Working Hours=300 Taka)

Total Developer Salary=300*1408*3 = **1267200**Taka

Requirement analysis

Time needed: 1 month (22 Working Days=176 Working Hour)

Req analysis person hourly wage: 250 Taka

Total Req Analysis Expense: 250*176 = **44,000** Taka

Transportation cost estimation: **9000** Taka

Training & Hardware Expanses Estimation: **60,000** Taka.

Rent Expenses:

Room per month: 7000 Taka

Total in 8 months: **56,000** Taka

Total utilities in 8 months: **7000** Taka

Maintenance (Till 5 Months after Delivery)

Expense per hour: 700 Taka

Total Estimated Time Needed for Maintenance = 40 Hours

Total Estimated Maintenance Expense =40*700=**28000** Taka

Total Estimated Expense:

1267200 + 44,000 + 9000 + 60,000 + 56,000 + 7000 + 28,000

= 1471200 Taka

Profit:

20% of Total Estimated Expense =294240 Taka

Project Budget: 1471200+294240=1765440 Taka

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Conclusion:

Road rescue app stands as a reliable companion for all road travellers. This project proposal outlines the app's features, target audience, technology, diagrams, and cost, aiming to create a solution that benefits the users with confidence and peace of mind during journeys. From promptly connecting users with essential roadside aid to guiding them to nearby hotels, hospitals, and gas stations, the app redefines convenience and safety on the road.